

Discussion III



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Problem Set



1. HW Example: Taxation in the 2-Period Model

Assume a person has a 2 period Utility function: $U = \ln(C_1) + 2 \ln(C_2)$. she has an income in the first period of $Y_1 = 100.000\text{€}$ and saves $S = Y_1 - C_1$, to finance consumption in the second period (no labor income in period 2; $Y_2 = 0$). The interest rate is $r = 10\%$.

a) Find the optimal allocation of income for consumption in periods 1 and 2 and the saved amount S of the person.

1. step: Budget constraint : $C_1 + \frac{C_2}{(1+r)} \leq Y_1 + \frac{Y_2}{(1+r)}$

$$C_1 + \frac{C_2}{(1,1)} \leq 100.000 + \frac{0}{(1,1)}$$

$$\frac{C_2}{(1,1)} \leq (100.000 - C_1)$$

$$C_2 \leq 110.000 - 1,1C_1$$

2. step : Substitute the budget constraint into the utility function

$$U = \ln(C_1) + 2 \ln(110.000 - 1,1C_1)$$

Differentiate wrt C_1 :

$$\frac{\partial U}{\partial C_1} = \frac{1}{C_1} - \frac{2,2}{110.000 - 1,1C_1} = 0$$

$$\rightarrow C_1 = 33.333,33$$

$$\rightarrow C_2 = 110.000 - 1,1C_1 = 73.333,33$$

$$\rightarrow S = Y_1 - C_1 = 66.666,66$$

b) The state introduces an income tax of 25%. How do consumption and savings change compared to a)?

b) 1. New Budget Constraint: $C_1 + \frac{C_2}{(1+r)} \leq Y_1(1 - \tau_Y) + \frac{Y_2}{(1+r)}(1 - \tau_Y)$

$$C_1 + \frac{C_2}{(1,1)} \leq 100.000(1 - 0,25) + \frac{0}{(1,1)}(1 - 0,25)$$

Smaller constant
than in a) → lower
income

$$C_2 \leq 82.500 - 1,1 C_1$$

Slope remains the
same! No substitution
effect

2. Substitute into Utility Function:

$$U = \ln(C_1) + 2 \ln(82.500 - 1,1C_1)$$

Differentiate wrt C_1 $\frac{\partial U}{\partial C_1} = \frac{1}{C_1} - \frac{2,2}{82.500 - 1,1C_1} = 0$

Calculated with
income after taxes!

$$\rightarrow C_1 = 25.000$$

$$\rightarrow S = Y_1(1 - 0,25) - C_1 = 50.000$$

$$\rightarrow C_2 = 82.500 - 1,1C_1 = 55.000$$

Savings decrease → Income effect
dominates

c) Which consumption tax rate has the same effect as the income tax of 25%?

1. Goal: New Budget Constraint: $(1 + \tau_{Cs})C_1 + (1 + \tau_{Cs})\frac{C_2}{(1+r)} \leq Y_1 + \frac{Y_2}{(1+r)}$

$$\Rightarrow (1 + \tau_{Cs}) \left\{ C_1 + \frac{C_2}{(1+r)} \right\} \leq Y_1 + \frac{Y_2}{(1+r)}$$

My consumption
becomes more
expensive, that's why
 $+ \tau_{Cs}$

2. From Income Tax (EKS): Budget Constraint: $C_1 + \frac{C_2}{(1+r)} \leq Y_1(1 - \tau_Y) + \frac{Y_2}{(1+r)}(1 - \tau_Y)$

$$\begin{aligned} \Rightarrow C_1 + \frac{C_2}{(1+r)} &\leq \left\{ Y_1 + \frac{Y_2}{(1+r)} \right\} (1 - \tau_Y) \\ \Rightarrow \frac{1}{(1 - \tau_Y)} \left\{ C_1 + \frac{C_2}{(1+r)} \right\} &\leq Y_1 + \frac{Y_2}{(1+r)} \end{aligned}$$

3. Equating:

$$\begin{aligned} (1 + \tau_{Cs}) &= \frac{1}{(1 - \tau_Y)} \\ \Rightarrow \tau_{Cs} &= \frac{1}{(1 - 0.25)} - 1 \\ \Rightarrow &\mathbf{33,33\%} \end{aligned}$$

Tax Avoidance and Evasion



Tax Avoidance/Evasion – A Trivial Offense?

- Blurred line between Avoidance and Evasion

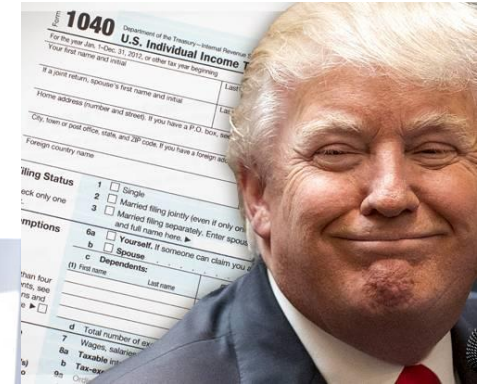


NACHRICHTEN MEINUNG MAGAZIN Österreich International Geld & Finanzen

Karl Heinz Grasser endgültig freigesprochen



Karl Heinz Grasser und sein Anwalt Norbert Wess vor Verhandlungsbeginn am Straflandesgericht Wien. Der Vorwurf der Steuerhinterziehung ist nun endgültig vom Tisch. APA/Roland Schlager



“Particularly cautious bank customers simply took the ski lifts from Oberstdorf and skied/commuted across with backpacks full of cash into the tranquil Kleinwalsertal.”

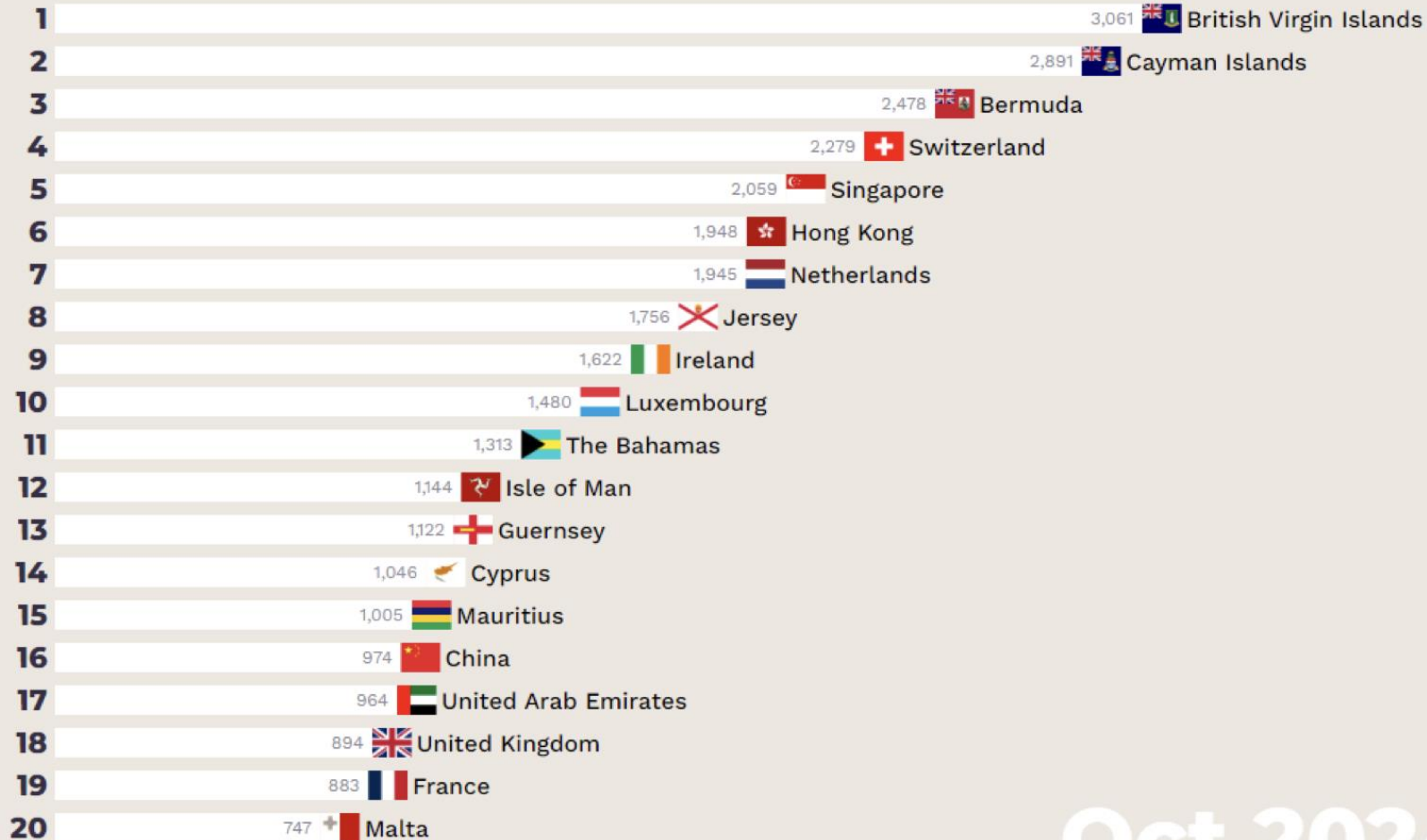


- **The Austrian tax haven for Germans. (6 billion)**
- **Until 2015: Softening of strict banking secrecy laws**

Which countries worldwide enable the most tax evasion (in absolute terms) by companies and private individuals?

Estimates from Tax Justice Network (2024)

Extent (Evasion and Avoidance)



Oct 2024

CHAFTS
RSITÄT
/IENNA
SITY OF
OMICS
BUSINESS

Legend:

- In Favour
- Abstention
- Against

Australia



These hurtful eight

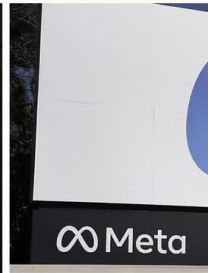
the lose-lose global

Share of global tax

The Guardian view on a tax war: the world must unite against American obstruction

Editorial

Donald Trump's threats to global tax reform have backfired, leaving the US isolated as nations push ahead with a new UN tax convention



Under discussion is a new UN tax convention that may permit states to tax economic activity where it actually occurs, rather than allowing multinationals to shift profits to tax havens. The Tax Justice Network (TJN) said last year that nations lose \$492bn (£390bn) annually due to corporate tax abuse. The global south bears the greatest losses, which undermine public services like health and education. If enacted, the convention would create a legally binding framework requiring multinationals to pay tax where they employ staff and do real business - not where they stash profits. This would replace the outdated arm's-length principle with unitary taxation, ensuring fair profit allocation. It would mean an end to Amazon, Google and

U.S. President Donald Trump's decisions have triggered significant shifts in international politics with ripple effects extending even to niche areas like corporate taxation. The U.S. has effectively withdrawn from global tax cooperation, jeopardising a decade of painstaking negotiations and casting uncertainty over the future of tax multilateralism. This marks the end of an era, raising pressing questions about the viability of global tax frameworks. However, it also presents an opportunity for countries of the so-called Global South to explore alternative approaches that may redefine international tax governance in the years to come.

Paper-Discussion



Bott, Cappelen, Sørensen, & Tungodden, 2020

- But before that, a small **Recap...**

- D_i : Indicator for unit i receiving the treatment.
- $D_i = 1$ for the Treatment Group
- $D_i = 0$ for the control Group

- **Potential outcomes:**
 - $Y_i(1)$ = Potential outcome for person with treatment i
 - $Y_i(0)$ = Potential outcome for person without treatment i

- **How can we define our causal effect:**
 - $Y_i(1) - Y_i(0)$: Where is the Problem?
 - „Fundamental Problem of Causal Inference“
 - We cannot observe both outcomes for person i !
 - Also average treatment effect (ATE) = $E[Y_i(1) - Y_i(0)]$

Too good to be true?

- $ATE = E [Y_i(1) - Y_i(0)]$
 - Not calculable, but **estimable** – why?
 - *"Difference in Means Estimator"*
- Which assumptions must be met?

$$\underbrace{\frac{1}{N_T} \sum_{i=1}^n (y_i | d_i = 1) - \frac{1}{N_C} \sum_{i=1}^n (y_i | d_i = 0)}_{\text{Simple Difference in Outcomes}} = \underbrace{E[Y^1] - E[Y^0]}_{\text{Average Treatment Effect}} + \underbrace{E[Y^0 | D = 1] - E[Y^0 | D = 0]}_{\text{Selection bias}} + \underbrace{(1 - \pi)(ATT - ATU)}_{\text{Heterogeneous treatment effect bias}}$$

1. Individuals were selected **Independently**

- Randomly assigned to control/treatment group (RCT vs. Observational)
- Important due to selection effects!

2. **Stable Unit Treatment Value Assumption (SUTVA)**

- Every treatment has the same effect
- No externalities ("Spillovers")
- Important due to heterogeneous treatment effects!

3. Sample is large enough ("**Law of Large Numbers**")

- **Natural Experiment:**

- An external factor, such as a **change in policy**, a **natural disaster**, or a **historical event**, causes a random or **quasi-random assignment** of treatment and control groups.
 - **E.g.:** Higher minimum wage in Carinthia (**Kärnten**)!

- **Assumptions:**

- No time-varying, **specific shocks** to the outcome variable that are correlated with the treatment status.
- **E.g.:** Carinthia must not simultaneously introduce a subsidy for companies
- Selection into the treatment is **independent**.
- **E.g.:** People from Styria (**Steierer**) must not identify the change in advance and relocate..

- **Hauptvorteil:**
 - **Randomisierung = zufällige Zuteilung zum Treatment.**
 - Später mehr dazu..

- Welche Arten von kontrollierten Experimenten kennt ihr?
 - **RCT:** Randomisiertes Feldexperiment
 - **Survey-Experiment**
 - **Lab Experiment**

Discussion in groups:

- **Group 1:** What are the advantages and disadvantages of the Randomized Control Trial (RCT) method for economic research? Were all assumptions fulfilled in the following paper?
- **Group 2:** Explain how the theoretical model works.
- **Group 3:** Explain results from Figures 1 and 3.
- **Group 4:** What do the long-term effects look like? Why? How would you critique these results?

What are the advantages and disadvantages of the Randomized Control Trial (RCT) method for economic research?

- **Advantages:**
 - Clean causal effect interpretations!
 - Confounding factors can be very well controlled ("**Observing the Unobservables!**")
 - Research questions that are otherwise difficult to answer can be investigated!
 - Very transparent from a research theory perspective (Pre-analysis plan, ...)
 - Closeness to practice leads to clearer and more direct impacts on policy, institutions, etc.
 - "**Iterative Experimentation!**"
- **Disadvantages:**
 - Often a very **small scope** (external validity)!
 - Mostly only very **short-term effects** ("long-term")
 - **Spillover effects!**
 - "**Attrition in the field**"
 - Experimenter effect: different behavior when one knows they are part of an experiment
 - Institutional, structural context is often neglected (in design, but especially in interpreting results)
 - Only applicable to very specific research questions ("small questions instead of large questions")
 - Ethical questions

[HTTPS://VOXEU.ORG/ARTICLE/WHAT-RANDOMISATION-CAN-AND-CANNOT-DO-2019-NOBEL-PRIZE](https://voxeu.org/article/what-randomisation-can-and-cannot-do-2019-nobel-prize)

[HTTPS://WWW.VOX.COM/FUTURE-PERFECT/2019/12/11/20938915/NOBEL-PRIZE-ECONOMICS-BANERJEE-DUFLO-KREMER-RCTS](https://www.vox.com/future-perfect/2019/12/11/20938915/nobel-prize-economics-banerjee-duflo-kremer-rcts)

- **Randomized?**
 - Smaller numbers, however, had a higher probability of being selected.
 - Why?
 - Poorly mixed! → **Uneven Distribution**
- **SUTVA (Stable Unit Treatment Value Assumption):**
 - No interference between treatments
 - Do vaccinations affect healthy people? (indirectly)
 - No spillovers! Can educational measures also influence others? (directly)
 - **No Treatment Heterogeneity:** Vaccinations work equally well for everyone!
- **Dropout-Rates are equal! (Independence Assumption)**
- Sufficiently large Sample (**Power!**)

You've Got Mail: A Randomized Field Experiment on Tax Evasion

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- Field experiment with 15,000 Norwegians (high probability of incorrectly reported foreign income)
- Foreign income must be self-reported
- International cooperation improving over the years
- Letters were sent shortly before taxes were due.
- Dependent Variable: Reported foreign income
- For the current and next year (= “long-run”)

Tax Morale: Non-financial reasons for tax compliance

Specific Assumptions:

- The **taxpayers were not aware** that NTA had received reports about their foreign income, and the tax authorities did not act on this information until after the taxpayers had self-reported foreign income for 2012...
- **Which assumption is violated?**
- **Selection effect!**

- “The moral letter“:
 - **Societal benefit:** “The great majority report information about their income and assets in Norway correctly and completely. In order to treat all taxpayers fairly, it is therefore important that foreign income and foreign assets are reported in the same manner.”
 - **Fairness:** “Your tax payment contributes to the funding of publicly financed services in education, health and other important sectors of society.”
- „Detection letter“:
 - “The tax administration has received information that you in previous years have had income and/or assets abroad.”
- Base Letter: Information (Foreign income must be reported, such as ...)

Balancing + Additional Stats

Table 5. Baseline Characteristics by Treatment

Treatment	<i>n</i>	Positive amount self-reported for 2011	Amount self-reported for 2011	Amount in ACA-reports for 2011	Amount in ACA-reports for 2012	Share female	Share above 60 years old	Share Norwegian citizen	High socioeconomic status
<i>No letter</i>	1,968	0.393 (0.011)	30,287 (4,356)	83,995 (33,390)	43,303 (6,761)	0.460 (0.011)	0.544 (0.011)	0.553 (0.011)	0.441 (0.011)
<i>Base</i>	3,947	0.402 (0.008)	27,427 (3,278)	55,211 (3,935)	44,104 (5,632)	0.450 (0.006)	0.551 (0.008)	0.553 (0.008)	0.454 (0.008)
<i>Moral</i>	7,821	0.405 (0.006)	28,040 (2,784)	51,971 (2,070)	38,097 (2,637)	0.459 (0.06)	0.561 (0.006)	0.548 (0.005)	0.454 (0.006)
<i>Detection</i>	1,972	0.417 (0.011)	24,689 (3,436)	47,855 (2,777)	58,313 (20,441)	0.446 (0.011)	0.542 (0.011)	0.548 (0.011)	0.452 (0.011)
Total	15,708	0.404 (0.004)	27,747 (1,756)	56,280 (4,434)	42,796 (3,321)	0.455 (0.004)	0.554 (0.004)	0.550 (0.004)	0.452 (0.004)

Notes. The table reports the baseline characteristics used as controls in the main regression specifications, by treatment. Standard errors of the means are in parentheses.

Table 3. Statistics from the Contact Logs

Statistic	Total	Treatment group			Test of equality <i>p</i> -values	
		<i>Base</i>	<i>Moral</i>	<i>Detection</i>	<i>Moral</i>	<i>Detection</i>
Share of letter recipients in the logs	0.050	0.036	0.037	0.131	0.710	<0.001
<i>Communication with letter recipient:</i>						
Question about why he/she receives letter	0.282	0.261	0.254	0.324	0.888	0.182
Negative reaction to letter	0.055	0.049	0.041	0.073	0.701	0.364
Question about rules of taxation	0.347	0.348	0.351	0.343	0.951	0.936
Arguing about legitimacy of rules	0.063	0.070	0.058	0.065	0.621	0.832
Question about how to report foreign income	0.721	0.741	0.758	0.668	0.709	0.126
Question about how foreign income is audited	0.019	0.007	0.014	0.031	0.540	0.127
Question about reporting income from before 2012	0.062	0.035	0.041	0.099	0.762	0.021
<i>Characteristics of caller:</i>						
Mean age	67.3	70.4	68.3	64.4	0.105	<0.001
Share female	0.544	0.643	0.558	0.475	0.089	<0.001
Share Norwegian citizen	0.611	0.664	0.605	0.589	0.223	0.138
Observations	700	143	294	263		

Notes. The table shows the share of taxpayers that contacted the call center, the share that asked each of the questions, and background characteristics. The table is based on 700 contacts with the call center in which callers identified which letter they had received. “Test of equality” reports a test of equality between the respective treatment and the *Base* treatment. The *p*-values are based on Pearson χ^2 tests on binary outcomes (and *t*-tests for tests of mean age).

Simple Model

$$EU(r; \cdot) = pu(Z) + (1 - p)u(Y) - s(T)\beta(y - r)^2,$$

y ... Foreign Income

r ... Reported Foreign Income

τ ... Penalty rate

p... Detection Probability



Detection T

Z: If incorreced reported and discovered= $y - t^*r - \tau(y - r)$

Y: If incorreced reported and not discovered = $y - t^*r$

s(T)... Salience of Moral Costs (Visibility)

β ... Moral Cost (if s(T) = 1, β full costs)



Moral T

$$r = y - \frac{\Delta u^s(r, t, \tau, p)}{2s\beta},$$



**Marginal Benefit /
Marginal Cost**

IF $s > 0$ und $\beta \rightarrow \infty$

$\Rightarrow r = y$

Selfish and Moral Types



Selfish and Moral Types

- **Selfish: $\beta = 0$** and **Moral: $\beta > 0$**

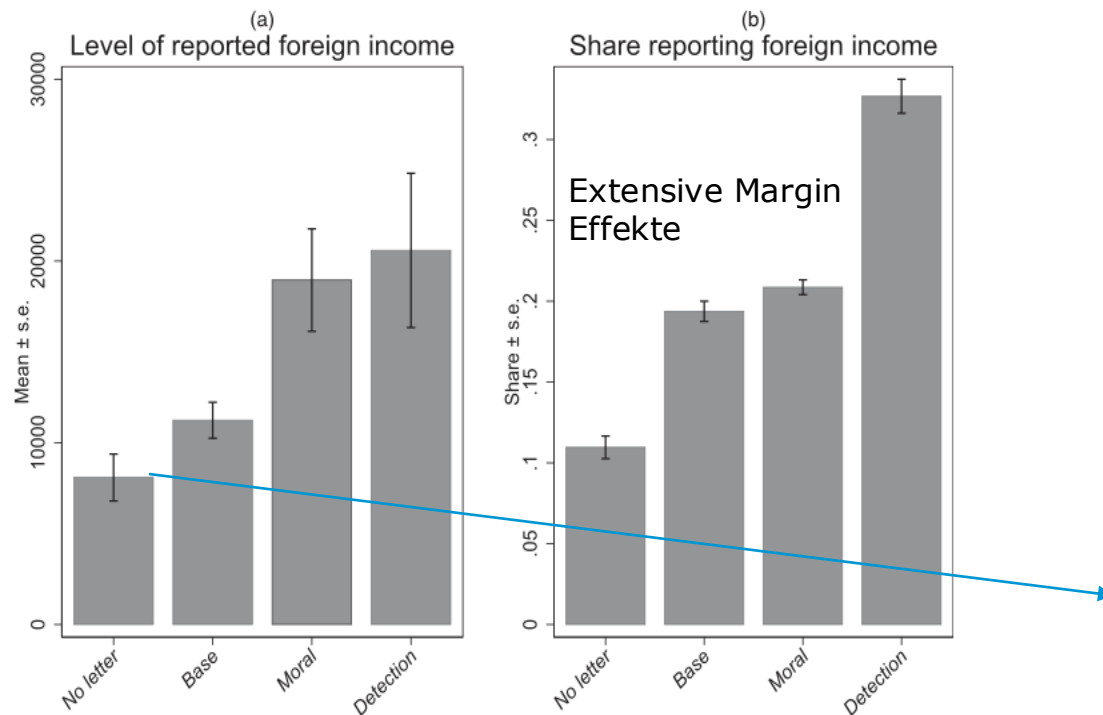
- **Moral Treatments only intensive Margin Effects:**
- **Change in $s(T)$ only affects moral types!**

- **Detection Treatment both:**
- **Changes in p have intensive and extensive effects!**
 - Selfish Types are also affected (Switching possible – **extensive**)
 - Moral & Selfish Types could also generally report more (**intensive**)

$$y_{i,t} = \alpha + \sum_{l \in L} \beta_l d_{il} + \delta y_{i,t_0} + \gamma \mathbf{x}_i + \varepsilon_i,$$

- $y_{i,t}$: Self-reported foreign income for individual i in Treatment year t .
- d_{il} : Indicator variable for which treatment individual i received..
- β_l : Estimated causal effect of treatment l compared to the reference treatment.
- y_{i,t_0} : Self-reported foreign income for the **Base Year** t_0 .
- $\mathbf{x}_i$: Control variables (Age, Gender, Norwegian citizenship, and a socio-economic status indicator, based on income and wealth).
- **With the "Monotone Treatment Response Assumption": Increased intensity of the treatment → no reduction in effects!**
 - Can calculate intensive/extensive margin effects!

Figure 1. Self-Reported Foreign Income

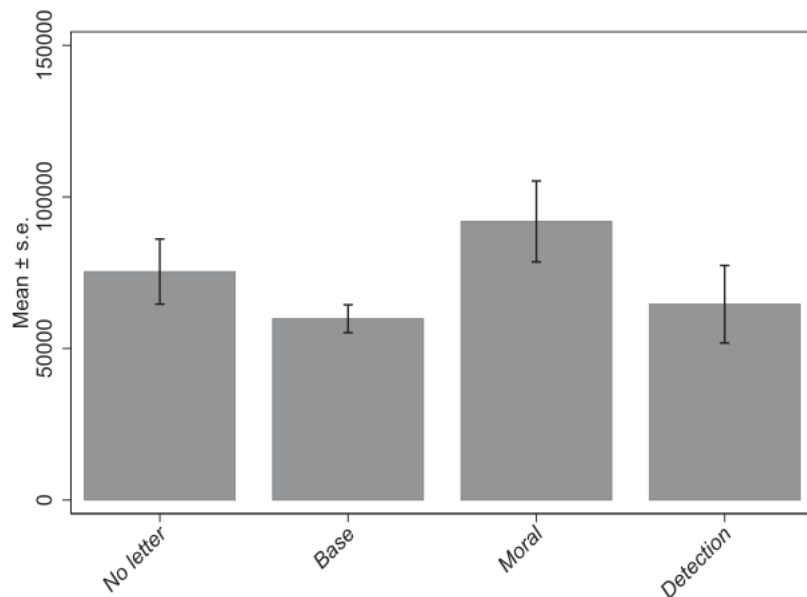


Note. The figure shows, for the follow-up year, the average amount of self-reported foreign income in NOK (panel (a)) and the share of taxpayers who self-report a positive foreign income (panel (b)), by treatment.

- **Large Effects:** From EUR 690 (without letter) to EUR 1,470, in the Treatment Groups
- 70% more Reporting with Moral and 80% with Detection
- Information gap cannot fully explain the willingness to report taxes

Effects for those who already reported positively:

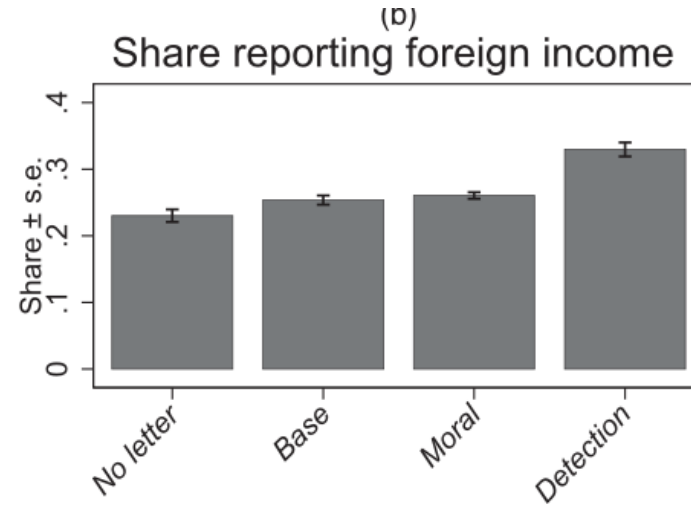
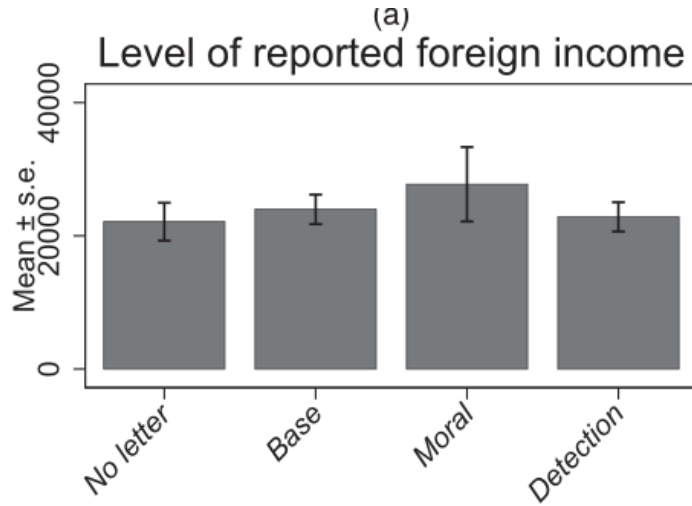
Figure 3. Self-Reported Foreign Income, Conditional on It Being Positive



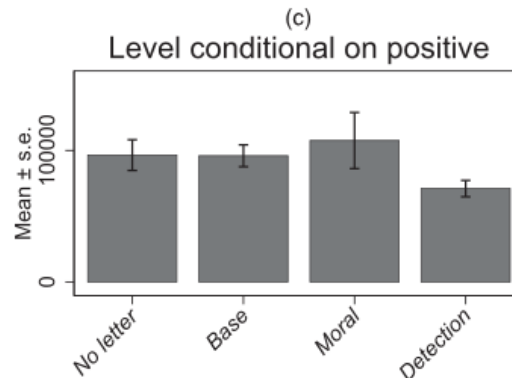
Note: The figure shows the average amount of self-reported foreign income in NOK for the subset of taxpayers that report a positive foreign income, by treatment.

- Which margin do we see here?
- Positive Level of reported Income:
 - Generally Intensive and Extensive.
 - But for the Moral Treatment no effect on the extensive Margin, so here only intensive Margin Effects!
 - For the Detection Letter no clue, because it includes large extensive Margin effects!!

„Long-Term“



- From 25% to 32%



Lower:

- „Long-Term“ = 1 year!

- Tax consulting
- Nothing happened last year?

- **End-term:**
- You should know the homework examples (**HÜ-Beispiele**)!
- You should understand the discussion paper.
 - Especially what we discuss in the unit about it.
- You do not need to understand the studies we often discuss as "**Empirical Evidence**" in detail.
 - They serve you as argumentation.
- You should be able to understand and apply the other models/theories we discuss.